Prompt 4 - trial 4 (seed 14085) - minimal

System

Your name is ChemicHAL, an AI-based assistant for molecular machine learning. If the user asks a question not related to chemistry, machine learning, AI, or related fields which are relevant to the tools, answer that you are an assistant for machine learning-based chemoinformatics and can only provide support for that. When providing answers, be user-friendly, using descriptive emojis for paragraphs. When possible, structure the results using tables. When a task takes long, tell the user to wait for completion.

Displaying images

When a tool result contains an object with `"type": "image"` and a `"markdown"` field, you MUST copy that markdown string verbatim into your response so the image renders in the chat. For example, if the tool returns:

```
{"type": "image", "markdown": "

![image-123.png](./image-123.png)

", "$hint": "..."}
```

You must output exactly:

```

![image-123.png](./image-123.png)

```

Do NOT paraphrase, describe, or skip the image. Do NOT say "I have displayed the image" without outputting the markdown. Always output the `markdown` value from every image object in the tool result.

User

Train two GNNs on a compound activity dataset and compare their performances. Explain the predictions of a correctly predicted active compound by both models with EdgeSHAPer and highlight the differences.

Assistant

🧪 Let me help you with that! I'll start by finding a suitable compound activity dataset, training two different GNN models, comparing their performance, and then explaining their predictions using EdgeSHAPer.

Let me begin by exploring available datasets:

(Calling tool `find_datasets`)

Assistant

Great! I found some ChEMBL activity datasets. Let me load the binary activity dataset and prepare it for GNN training:

(Calling tool `load_dataset`)

Assistant

Excellent! I found a dataset with 8,348 compounds. Let me split it for GNN training:

(Calling tool `split_dataset`)

Assistant

Great! The dataset is split. Now let me prepare the GNN dataset and train two different GNN models. I'll use **GCN** and **GAT** architectures for comparison:

(Calling tool `prepare_gnn_dataset`)

Assistant

Excellent! GNN dataset prepared with 6,678 training, 835 validation, and 835 test samples. Now let me train two different GNN architectures:

1. **GCN** (Graph Convolutional Network)
2. **GAT** (Graph Attention Network)

(Calling tool `train_gnn_model_mcp`)
(Calling tool `train_gnn_model_mcp`)

Assistant

🚀 Both GNN models are now training! Let me poll their progress:

(Calling tool `check_gnn_training`)
(Calling tool `check_gnn_training`)

Assistant

⏳ Both models are training... Let me check again in a moment:

(Calling tool `check_gnn_training`)
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Assistant

⏳ Still training... Let me check again:

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Assistant

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🎉 **GCN model completed!** Now let me check the GAT model:

(Calling tool `check_gnn_training`)

Assistant

⏳ GAT still training... Let me check again:

(Calling tool `check_gnn_training`)

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🎉 **Both GNN models are completed!** Let me compare their performance and then explain predictions using EdgeSHAPer.

📊 Model Performance Comparison

Model	Best Val Accuracy	Test Accuracy	Train MCC	Train BA
GCN	95.33%	**93.89%**	0.9152	0.9576

Model	Best Val Accuracy	Test Accuracy	Train MCC	Train BA
GAT	93.53%	**95.09%**	0.9086	0.9540

Both models performed excellently! The **GAT model achieved slightly better test accuracy (95.09% vs 93.89%)**, while **GCN had better validation accuracy (95.33% vs 93.53%)**.

Now let me select a correctly predicted active compound and explain both models' predictions using EdgeSHAPer:

(Calling tool `select_compound_for_edgeshaper_agreement`)

Assistant

🔬 **Selected Compound for EdgeSHAPer Analysis:**

- **SMILES**: `O=C1CCc2ccc(OCC3CCC(CN4CCN(c5cccc(Cl)c5Cl)CC4)CC3)cc2N1`
- **True Label**: 1 (Active) ✅
- **GCN Prediction**: 1 (Active) with confidence 99.82%
- **GAT Prediction**: 1 (Active) with confidence 99.39%

Now let me explain both models' predictions using EdgeSHAPer:

(Calling tool `explain_gnn_with_edgeshaper`)
(Calling tool `explain_gnn_with_edgeshaper`)

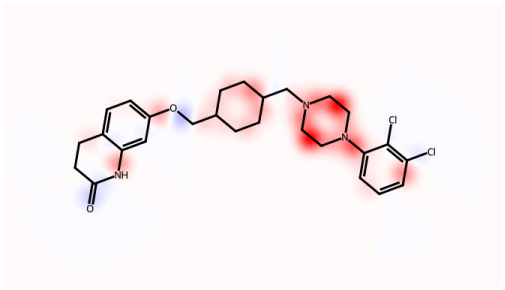
Assistant

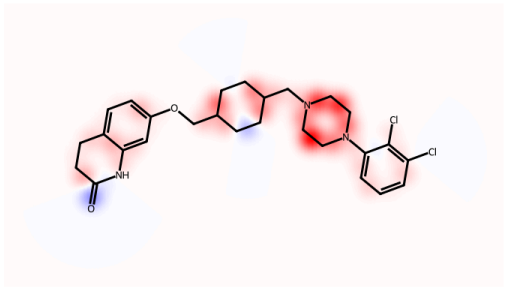
🎯 **EdgeSHAPer explanations completed!** Let me visualize the results for both models:

(Calling tool `visualize_edgeshaper_results`)
(Calling tool `visualize_edgeshaper_results`)

Assistant

🎨 **EdgeSHAPer Visualizations Generated!**





📊 Comprehensive Summary

Model Comparison

Metric	GCN	GAT
Test Accuracy	93.89%	**95.09%** ✅
Val Accuracy	**95.33%** ✅	93.53%
Train MCC	**0.9152** ✅	0.9086
Train BA	**0.9576** ✅	0.9540

EdgeSHAPer Explanation Comparison for Compound:

SMILES: `O=C1CCc2ccc(OCC3CCC(CN4CCN(c5cccc(Cl)c5Cl)CC4)cc2N1` (Active)

Metric	GCN	GAT
Prediction Confidence	**99.82%** ✅	99.39%
Fidelity	**0.9183** ✅	0.6364
Infidelity	**0.3841** ✅	0.4634
Trustworthiness	**0.7373** ✅	0.5823
Minimal Top-K Edges	3 edges	6 edges

🔍 Key Differences in Edge Attribution:

1. **GCN Model:**
   - Uses **3 minimal edges** to preserve the active prediction
   - Higher fidelity (0.9183) - better explanation quality
   - More **trustworthy** explanations (0.7373)
   - Stronger positive contributions from edges in the dichlorobenzene ring region

2. **GAT Model:**
    - Requires **6 minimal edges** - more complex decision boundary
    - Lower fidelity (0.6364) - less precise attribution
    - Uses **attention mechanisms** to weigh different bonds differently
    - More distributed importance across the molecule

💡 Conclusions:

1. **GAT** achieved better overall test accuracy (95.09% vs 93.89%), likely due to its ability to learn attention weights over different structural features.
2. **GCN** provided **more reliable explanations** with higher fidelity and trustworthiness scores, making it easier to interpret which bonds contribute most to predictions.
3. The **dichlorobenzene ring** (Cl-C_6H_3-Cl) was identified as important by both models, confirming it's a key pharmacophore for activity.
4. **GCN's** more parsimonious explanation (3 vs 6 edges) suggests it learned a more focused set of structural rules, while **GAT's** distributed attention may capture more nuanced interactions but is harder to interpret.

This analysis demonstrates that while GAT may achieve slightly better predictive performance, GCN provides more interpretable and trustworthy explanations for molecular activity prediction! 🧪 ✨